Canonical Nonlinear Global-Chart Structural Decomposition in the Penalized Quadratic Regime

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Abstract

In this work, I formulate the nonlinear global-chart structural decomposition of evolving densities in the first regime beyond the affine theory. Instead of representing the density only through mass, translation, a global affine deformation, and a normalized residual profile, I represent it through mass, a nonlinear global structural chart, and a normalized reference profile. I keep the affine theory as the exact low-order core, define a hybrid gauge in which the affine block is fixed canonically and the nonlinear block is selected variationally, and close the theory locally first in smooth admissible Banach families, with finite-dimensional families and the penalized quadratic class as explicit realizations. In that setting, I prove local identifiability from positivity of the reduced Hessian and local robustness from the implicit function theorem, and I state the corresponding Banach-space extension under the usual Frechet differentiability and isomorphism hypotheses. I also introduce a variational reaction-transport action, derive the associated structural continuity equation with source, and define the corresponding structural distance together with its formal dual form. Finally, I record both the classical multiseed comparison against the affine theory and a degraded observational validation, together with a multichart proxy, as computational evidence for the single-chart nonlinear regime and for its ontological limit.

1 Introduction

My goal is to describe density evolution in coordinates that separate the affine structural core from genuinely nonlinear single-chart deformation. In the affine theory, the density is decomposed into mass, translation, a canonical affine deformation, and a normalized residual shape. That regime already provides a clean structural language, as established in the affine structural manuscript [1], but the accumulated experiments show that some densities remain globally coherent while no longer being adequately described by a single affine chart.

The central question in the present manuscript is therefore the following: how far can one push the single-chart ontology before one must change language and move to a multichart regime? The answer proposed here is the regime $V3$, understood as the canonical nonlinear global-chart structural decomposition. I do not attempt here to close the full nonlinear theory in complete generality. Instead, I close its local mathematical core first in smooth admissible Banach families and make that closure explicit in finite-dimensional subclasses, beginning with the penalized quadratic class.

The affine theory remains structurally fundamental. The present work does not replace it. Rather, it treats the affine decomposition as the exact low-order core of a broader single-chart theory. The multichart theory likewise remains outside the scope of the present manuscript; it enters only when the single-chart ontology ceases to be structurally adequate. In addition, the present version equips $V3$ with an explicit reaction-transport variational principle, so that the regime is not only kinematic but also endowed with a natural dynamic cost and an associated structural distance.

The closest external comparison is with unbalanced optimal transport and reaction-transport geometries such as Wasserstein-Fisher-Rao, Hellinger-Kantorovich, and entropy-transport formulations [2, 3, 4]. The present manuscript does not compete with those theories at the level of unrestricted metric generality. Its scope is different: I restrict the admissible class to structurally meaningful single-chart paths and seek a canonical reduced description with explicit affine anchoring, nonlinear chart selection, and an ontological transition criterion toward a multichart regime. In that sense, $V3$ should be read as a structural reduced theory of single-chart reaction-transport rather than as a replacement for the full unbalanced optimal transport state of the art.

2 Nonlinear Structural Variables

Let $\rho(x) \geq 0$ be a density on $\mathbb{R}^d$ with finite mass and finite second moment. Fix a symmetric positive definite reference matrix $\Sigma_0 \in \mathbb{R}^{d\times d}$.

Definition 1 (Nonlinear structural variables)**.** *I define the total mass*

$$M = \int_{\mathbb{R}^d} \rho(x)\, dx$$

and the barycenter

$$q = \frac{1}{M} \int_{\mathbb{R}^d} x\, \rho(x)\, dx.$$

I also define the centered covariance matrix

$$C_\rho = \frac{1}{M} \int_{\mathbb{R}^d} (x-q) \otimes (x-q)\, \rho(x)\, dx.$$

A nonlinear structural chart is a map of the form

$$\Phi(\xi) = q + A\xi + N(\xi),$$

where $A \in \mathbb{R}^{d\times d}$ is symmetric positive definite and N is a nonlinear correction.

Definition 2 (Canonical affine anchor)**.** *I call the affine block canonical when A is the unique symmetric positive definite matrix satisfying*

$$A\Sigma_0 A^\top = C_\rho.$$

Equivalently,

$$A = \Sigma_0^{-1/2} \big(\Sigma_0^{1/2} C_\rho \Sigma_0^{1/2}\big)^{1/2} \Sigma_0^{-1/2}.$$

This preserves the affine theory of [1] as the exact low-order anchor of the nonlinear theory.

Definition 3 (Gaussian reference profile)**.** *Given the structural covariance target Σ_0, I define the centered Gaussian reference profile by*

$$G_{\Sigma_0}(\xi) := \frac{1}{(2\pi)^{d/2} \det(\Sigma_0)^{1/2}} \exp\left(-\frac{1}{2}\xi^\top \Sigma_0^{-1} \xi\right).$$

The role of G_{Σ_0} in $V3$ is not to replace the structural residual profile η, but to provide a canonical benchmark inside the normalized class. In particular, it gives a preferred zero-order reference for measuring non-Gaussian residual structure after the affine anchor and the nonlinear chart have been fixed. At the present stage of the theory, Gaussianization is therefore a diagnostic normalization tool rather than an additional gauge constraint. Consequently, the official hybrid gauge of the present manuscript does not use structural Gaussianization as a preferential normalization. The gauge remains anchored by the affine normalization and by variational chart selection, while Gaussian comparison is kept as an auxiliary diagnostic layer.

3 Nonlinear Structural Decomposition

The nonlinear structural decomposition is based on the pushforward identity

$$\rho = M\,\Phi_{\#}\eta.$$

When Φ is admissible, this means

$$\rho(x) = \frac{M\,\eta(\xi)}{|\det D\Phi(\xi)|}, \qquad x = \Phi(\xi).$$

Definition 4 (Admissible chart and Jacobian margin)**.** *I call a nonlinear structural chart $\Phi(\xi) = q + A\xi + N(\xi)$ admissible on a structural domain Ω if:*

- *Φ is a C^1 diffeomorphism from Ω onto its image $\Phi(\Omega)$;*
- *$\det D\Phi(\xi) > 0$ for every $\xi \in \Omega$;*
- *the residual profile induced by Φ, as defined below, is normalized.*

I say that Φ has Jacobian margin at least $\kappa > 0$ on Ω if

$$\det D\Phi(\xi) \geq \kappa \qquad \text{for every } \xi \in \Omega.$$

Definition 5 (Normalized nonlinear residual profile)**.** *Given a C^1 chart Φ with positive Jacobian on Ω, I define the residual profile by*

$$\eta(\xi) = \frac{1}{M}\rho(\Phi(\xi))|\det D\Phi(\xi)|.$$

I call η normalized when

$$\int_\Omega \eta(\xi)\,d\xi = 1, \qquad \int_\Omega \xi\,\eta(\xi)\,d\xi = 0, \qquad \int_\Omega \xi\otimes\xi\,\eta(\xi)\,d\xi = \Sigma_0.$$

Definition 6 (Hybrid gauge)**.** *I call the gauge hybrid when:*

- *the affine block (M, q, A) is fixed by the canonical affine normalization;*
- *the nonlinear block N is selected variationally inside an admissible class;*
- *the nonlinear block is constrained by*

$$N(0) = 0, \qquad DN(0) = 0,$$

$$\int N(\xi)\eta(\xi)\,d\xi = 0, \qquad \int \xi\otimes N(\xi)\eta(\xi)\,d\xi = 0.$$

The affine theory is recovered exactly when $N \equiv 0$.

Proposition 1 (Strong affine compatibility)**.** *Assume that the admissible nonlinear class contains the null correction $N \equiv 0$. Then the single-chart nonlinear theory contains the affine theory exactly: the affine chart $\Phi_0(\xi) = q + A\xi$ is admissible, the pulled-back nonlinear residual coincides with the affine residual profile, all nonlinear gauge constraints are satisfied trivially, and if the structural functional is minimized at $N \equiv 0$, then the canonical $V3$ decomposition coincides exactly with the canonical affine decomposition.*

Proof. Because $N \equiv 0$ belongs to the admissible class, the chart $\Phi_0(\xi) = q + A\xi$ is an admissible candidate of the nonlinear theory. Moreover, the anchoring conditions $N(0) = 0$, $DN(0) = 0$, $\int N(\xi)\eta(\xi)\,d\xi = 0$, and $\int \xi \otimes N(\xi)\eta(\xi)\,d\xi = 0$ are satisfied identically.

For this null correction,

$$D\Phi_0(\xi) = A, \qquad |\det D\Phi_0(\xi)| = \det A.$$

Hence the pulled-back residual profile of the nonlinear theory is

$$\eta_0(\xi) = \frac{1}{M}\rho(q + A\xi)\det A.$$

Using the affine decomposition,

$$\rho(x) = \frac{M}{\det A}\eta\big(A^{-1}(x - q)\big),$$

and substituting $x = q + A\xi$, I obtain

$$\eta_0(\xi) = \frac{1}{M} \cdot \frac{M}{\det A}\,\eta\big(A^{-1}(q + A\xi - q)\big)\det A = \eta(\xi).$$

Therefore the nonlinear residual associated with $N \equiv 0$ coincides exactly with the affine residual profile.

Finally, if the structural functional attains its minimum at $N \equiv 0$, then the hybrid gauge selects precisely the affine chart $\Phi_*(\xi) = q + A\xi$ and the same normalized residual profile $\eta_* = \eta$. Thus the canonical $V3$ decomposition is exactly the canonical affine decomposition. □

4 Variational Reaction-Transport Principle

The structural decomposition becomes dynamically meaningful once it is equipped with a reaction-transport action. I therefore consider time-dependent paths

$$\rho_t = M_t(\Phi_t)_\# \eta_t, \qquad t \in [0, 1],$$

where $M_t > 0$, Φ_t is an admissible chart, and η_t is a normalized structural profile.

Definition 7 (Structural reaction-transport action)**.** *Let $w_t(\xi)$ be an internal structural transport field and let $r_t(\xi)$ be an internal structural reaction field, so that*

$$\partial_t \eta_t + \nabla_\xi \cdot (\eta_t w_t) = \eta_t r_t$$

in structural coordinates. I define the associated Eulerian velocity and reaction rate by

$$v_t(\Phi_t(\xi)) = \partial_t \Phi_t(\xi) + D\Phi_t(\xi) w_t(\xi),$$

$$g_t(\Phi_t(\xi)) = \frac{\dot{M}_t}{M_t} + r_t(\xi).$$

For a reaction weight $\lambda > 0$, I define the structural action of the path by

$$\mathcal{A}_{V3}(M, \Phi, \eta, w, r) := \int_0^1 \int_{\Omega_t} \left(\frac{1}{2}|v_t(x)|^2 + \frac{\lambda}{2}|g_t(x)|^2 \right) \rho_t(x)\,dx\,dt.$$

Equivalently, in structural coordinates,

$$\mathcal{A}_{V3} = \int_0^1 \int_\Omega \left(\frac{1}{2}|\partial_t \Phi_t(\xi) + D\Phi_t(\xi) w_t(\xi)|^2 + \frac{\lambda}{2}\left| \frac{\dot{M}_t}{M_t} + r_t(\xi) \right|^2 \right) M_t \eta_t(\xi)\,d\xi\,dt.$$

Proposition 2 (Derived continuity equation with source)**.** *Assume that $(M_t, \Phi_t, \eta_t, w_t, r_t)$ is a smooth structural path in $V3$. Then the induced density $\rho_t = M_t(\Phi_t)_\#\eta_t$ satisfies the Eulerian reaction-transport equation*

$$\partial_t \rho_t + \nabla_x \cdot (\rho_t v_t) = \rho_t g_t.$$

Proof. Let $\psi \in C_c^\infty(\mathbb{R}^d)$ be a test function. By the definition of pushforward,

$$\int_{\mathbb{R}^d} \psi(x)\rho_t(x)\,dx = M_t \int_\Omega \psi(\Phi_t(\xi))\eta_t(\xi)\,d\xi.$$

Differentiating in time yields

$$\frac{d}{dt}\int \psi\,\rho_t = \dot{M}_t \int \psi(\Phi_t)\eta_t + M_t \int \nabla\psi(\Phi_t)\cdot \partial_t\Phi_t\,\eta_t + M_t \int \psi(\Phi_t)\,\partial_t\eta_t.$$

Using the structural balance law

$$\partial_t \eta_t + \nabla_\xi \cdot (\eta_t w_t) = \eta_t r_t,$$

I rewrite the last term as

$$\int \psi(\Phi_t)\partial_t\eta_t = \int \psi(\Phi_t)\eta_t r_t - \int \psi(\Phi_t)\nabla_\xi \cdot (\eta_t w_t).$$

After integration by parts in ξ, the transport term becomes

$$-\int \psi(\Phi_t)\nabla_\xi \cdot (\eta_t w_t) = \int \nabla_\xi(\psi\circ\Phi_t)\cdot w_t\,\eta_t = \int \nabla\psi(\Phi_t)\cdot D\Phi_t(\xi)w_t(\xi)\,\eta_t(\xi)$$

assuming the standard vanishing boundary contribution for the admissible path. Therefore,

$$\frac{d}{dt}\int \psi\,\rho_t = M_t \int \nabla\psi(\Phi_t)\cdot\big(\partial_t\Phi_t + D\Phi_t w_t\big)\eta_t + M_t \int \psi(\Phi_t)\left(\frac{\dot{M}_t}{M_t} + r_t\right)\eta_t.$$

By the definitions

$$v_t(\Phi_t(\xi)) = \partial_t\Phi_t(\xi) + D\Phi_t(\xi)w_t(\xi), \qquad g_t(\Phi_t(\xi)) = \frac{\dot{M}_t}{M_t} + r_t(\xi),$$

and pushing forward again to Eulerian coordinates, I obtain

$$\frac{d}{dt}\int_{\mathbb{R}^d} \psi(x)\rho_t(x)\,dx = \int_{\mathbb{R}^d} \nabla\psi(x)\cdot v_t(x)\rho_t(x)\,dx + \int_{\mathbb{R}^d} \psi(x)g_t(x)\rho_t(x)\,dx.$$

This is exactly the weak form of

$$\partial_t \rho_t + \nabla_x \cdot (\rho_t v_t) = \rho_t g_t.$$

□

Definition 8 (Structural distance)**.** *For two admissible densities ρ_0 and ρ_1 in the single-chart regime, I define the $V3$ structural reaction-transport distance by*

$$\mathsf{D}_{V3,\lambda}(\rho_0,\rho_1)^2 := \inf \mathcal{A}_{V3}(M,\Phi,\eta,w,r),$$

where the infimum is taken over all admissible structural paths joining ρ_0 to ρ_1.

Proposition 3 (Formal dual formulation)**.** *The natural dual candidate associated with* $\mathsf{D}_{V3,\lambda}$ *is obtained by maximizing endpoint pairings against scalar potentials* φ_t *satisfying the Hamilton-Jacobi-reaction inequality*

$$\partial_t\varphi_t + \frac{1}{2}|\nabla_x\varphi_t|^2 + \frac{1}{2\lambda}|\varphi_t|^2 \le 0.$$

In this sense, $V3$ *carries a structural analog of the Wasserstein-Fisher-Rao or Hellinger-Kantorovich duality, restricted to admissible single-chart paths.*

Proof. Let (ρ_t, v_t, g_t) be an admissible structural path joining ρ_0 to ρ_1, and let φ_t satisfy the Hamilton-Jacobi-reaction inequality. Using the continuity equation with source,

$$\partial_t\rho_t + \nabla_x \cdot (\rho_t v_t) = \rho_t g_t,$$

and testing against φ_t, I obtain the weak identity

$$\frac{d}{dt}\int \varphi_t\rho_t = \int \big(\partial_t\varphi_t + \nabla\varphi_t \cdot v_t + \varphi_t g_t\big)\rho_t.$$

Integrating in time from 0 to 1 gives

$$\int \varphi_1\rho_1 - \int \varphi_0\rho_0 = \int_0^1\int \big(\partial_t\varphi_t + \nabla\varphi_t \cdot v_t + \varphi_t g_t\big)\rho_t\,dx\,dt.$$

Now apply Young's inequality in the transport and reaction variables:

$$\nabla\varphi_t \cdot v_t \le \frac{1}{2}|\nabla\varphi_t|^2 + \frac{1}{2}|v_t|^2, \qquad \varphi_t g_t \le \frac{1}{2\lambda}|\varphi_t|^2 + \frac{\lambda}{2}|g_t|^2.$$

Therefore,

$$\partial_t\varphi_t + \nabla\varphi_t \cdot v_t + \varphi_t g_t \le \partial_t\varphi_t + \frac{1}{2}|\nabla\varphi_t|^2 + \frac{1}{2\lambda}|\varphi_t|^2 + \frac{1}{2}|v_t|^2 + \frac{\lambda}{2}|g_t|^2.$$

If φ_t satisfies the stated Hamilton-Jacobi-reaction inequality, the first three terms are nonpositive, so

$$\int \varphi_1\rho_1 - \int \varphi_0\rho_0 \le \int_0^1\int \left(\frac{1}{2}|v_t|^2 + \frac{\lambda}{2}|g_t|^2\right)\rho_t\,dx\,dt.$$

Taking the infimum over admissible structural paths yields the dual upper bound associated with $\mathsf{D}_{V3,\lambda}$. This is the formal Legendre-dual counterpart of the structural action, with the essential restriction that optimization is performed only over admissible single-chart paths. □

5 Functional Setting in Finite-Dimensional Admissible Families

To close the theory locally in a form that can also cover genuinely infinite-dimensional models, I first formulate the gauge on smooth admissible Banach families and then specialize to finite-dimensional admissible families.

Definition 9 (Finite-dimensional admissible family)**.** *I call a nonlinear family finite-dimensional and admissible when*

$$N_\theta(\xi) = \sum_{\alpha=1}^{m} \theta_\alpha h_\alpha(\xi), \qquad \theta \in \mathbb{R}^m,$$

with fixed basis fields h_α*, and when the associated charts remain admissible for* θ *in an open parameter set. I write the corresponding penalized structural functional as*

$$J_{\mathcal{F}}(\theta; R, \omega).$$

Definition 10 (Banach admissible family)**.** *I call a nonlinear family Banach-admissible when the nonlinear block belongs to a Banach space $\mathcal{Y}$ of admissible corrections, the structural targets belong to a Banach space $\mathcal{X}$, the gauge weights belong to a parameter space $\mathcal{W}$, and the stationarity map*

$$F_{\mathcal{Y}}(N; R, \omega) = D_N J_{\mathcal{Y}}(N; R, \omega)$$

is well defined on an open subset of $\mathcal{Y} \times \mathcal{X} \times \mathcal{W}$.

Theorem 1 (Banach-space extension of the local theory)**.** *Let N_* be a stationary admissible chart in a Banach-admissible family, with reference target R_0 and weights ω_0. Assume that:*

- *the functional $J_{\mathcal{Y}}$ is twice Frechet differentiable in N and continuously Frechet differentiable in (R, ω);*
- *the linearized stationarity operator*

$$D_N F_{\mathcal{Y}}(N_*; R_0, \omega_0)$$

  is a Banach-space isomorphism from $\mathcal{Y}$ onto its dual target space;
- *the admissible-chart condition is open in $\mathcal{Y}$;*
- *the second variation is coercive on the admissible tangent space at N_*.*

Then the gauge is locally identifiable at N_* and there exists a unique local C^1 branch

$$(R, \omega) \mapsto N(R, \omega)$$

of admissible stationary charts near (R_0, ω_0). In particular, the local robustness and local identifiability theory of $V3$ extends to the Banach family $\mathcal{Y}$.

Proof. Coercivity of the second variation yields local strict convexity on the admissible tangent space, hence local identifiability. The Banach-space implicit function theorem applies to the stationarity equation because the derivative of the stationarity map is assumed to be an isomorphism. Openness of the admissible-chart condition guarantees that the resulting branch remains inside the admissible regime. Therefore the same local uniqueness and local stability mechanism extends from finite-dimensional families to the Banach family $\mathcal{Y}$. □

Definition 11 (Stationarity map)**.** *I define the stationarity map*

$$F_{\mathcal{F}}(\theta; R, \omega) = D_\theta J_{\mathcal{F}}(\theta; R, \omega).$$

A parameter vector θ_* is called stationary if

$$F_{\mathcal{F}}(\theta_*; R, \omega) = 0.$$

Proposition 4 (Transfer of the local theory to finite-dimensional families)**.** *Assume that $J_{\mathcal{F}}$ is C^2 in θ, that the admissible-chart condition is open in parameter space, and that the reduced Hessian at a stationary point θ_* is positive definite. Then the local identifiability and local robustness arguments of $V3$ apply in the same way to the family $\mathcal{F}$.*

Proof. Once the nonlinear block is reduced to finitely many parameters, the local theory becomes a finite-dimensional variational problem. Positivity of the reduced Hessian gives local strict convexity, and invertibility of the derivative of the stationarity map allows one to invoke the implicit function theorem. Openness of the admissibility condition guarantees that the resulting branch stays inside the admissible-chart regime. Therefore the same local uniqueness and stability arguments used below for the quadratic class transfer to any smooth finite-dimensional admissible family. □

Definition 12 (Penalized quadratic class)**.** *In dimension $d = 2$, I write the nonlinear block as*

$$N_c(\xi) = \sum_{\alpha=1}^{m} c_\alpha h_\alpha(\xi),$$

where the basis functions h_α are centered quadratic features associated with

$$H_2(\xi) = \xi \otimes \xi - \Sigma_0.$$

I call this the penalized quadratic class when the coefficients c are selected by minimizing a penalized functional

$$J_{\text{quad}}(c; R, \omega),$$

combining reconstruction, third-moment mismatch, affine-orthogonality penalty, Jacobian penalty, and nonlinear regularization.

This quadratic realization is the first explicit subclass in which the manuscript writes all quantities in closed form and computes the corresponding chart numerically.

6 Local Identifiability

The nonlinear theory is not meaningful unless the selected chart is locally distinguishable from nearby admissible competitors.

Definition 13 (Local identifiability)**.** *I say that the gauge is locally identifiable at a stationary point c_* if there exists a neighborhood U of c_* such that every admissible stationary point of the restricted problem in U coincides with c_*. Equivalently, after the affine anchor and the admissibility constraints are fixed, no nontrivial nearby parameter produces the same local stationary chart.*

Theorem 2 (Local identifiability in the penalized quadratic class)**.** *Let c_* be a stationary point of $J_{\text{quad}}(\cdot; R_0, \omega_0)$ in the penalized quadratic class. Assume that:*

- *J_{quad} is C^2 in a neighborhood of c_*;*
- *the admissible-chart condition is open near c_*;*
- *the reduced Hessian*
$$H_* = D_c^2 J_{\text{quad}}(c_*; R_0, \omega_0)$$
is positive definite on the admissible tangent space;
- *the associated chart Φ_{c_*} preserves a positive Jacobian margin in that neighborhood.*

Then there exists $r > 0$ such that c_* is the unique admissible stationary point of $J_{\text{quad}}(\cdot; R_0, \omega_0)$ in the ball $B_r(c_*)$. In particular, the gauge is locally identifiable in the penalized quadratic class.

Proof. Because c_* is stationary, the first variation vanishes at c_*. Positivity of the reduced Hessian on the admissible tangent space implies strict local convexity of the restricted functional in a sufficiently small neighborhood of c_*. Hence there exists $r > 0$ such that every admissible perturbation h with $\|h\| < r$ satisfies

$$J_{\text{quad}}(c_* + h; R_0, \omega_0) > J_{\text{quad}}(c_*; R_0, \omega_0)$$

whenever $h \neq 0$ and $c_* + h$ remains admissible.

Suppose now that another admissible stationary point $\tilde{c} \in B_r(c_*)$ existed. Since both c_* and $\tilde{c}$ would belong to the same locally strictly convex admissible neighborhood, strict convexity would be violated. Therefore no such $\tilde{c}$ exists, and c_* is the unique admissible stationary point near itself.

The openness of the admissible-chart condition and the positive Jacobian margin guarantee that the neighborhood can be chosen entirely inside the admissible regime. Thus the uniqueness statement is genuinely a uniqueness statement for admissible charts, not merely for unconstrained parameters. This is precisely local identifiability. □

7 Local Robustness

Local robustness means that the selected chart varies continuously and uniquely under small perturbations of the structural target and of the gauge weights.

Theorem 3 (Local robustness in the penalized quadratic class)**.** *Let (c_*, R_0, ω_0) satisfy*

$$F(c_*; R_0, \omega_0) = 0.$$

Assume that:

- *J_{quad} is C^2 in c and C^1 in (R, ω);*
- *the admissible-chart condition is open near c_*;*
- *the reduced Hessian*
$$D_c F(c_*; R_0, \omega_0) = D_c^2 J_{\text{quad}}(c_*; R_0, \omega_0)$$
is invertible;
- *the chart Φ_{c_*} has uniformly positive Jacobian margin in a reference neighborhood.*

Then there exist neighborhoods $\mathcal{U}$ of (R_0, ω_0) and $\mathcal{V}$ of c_*, and a unique C^1 map

$$(R, \omega) \mapsto c(R, \omega) \in \mathcal{V}$$

such that

$$F(c(R, \omega); R, \omega) = 0$$

for all $(R, \omega) \in \mathcal{U}$. Moreover, after possibly shrinking $\mathcal{U}$, there exists $C > 0$ such that

$$\|c(R, \omega) - c_*\| \leq C\big(\|R - R_0\| + \|\omega - \omega_0\|\big).$$

In particular, the canonical quadratic chart depends locally in a unique and stable way on the structural target and on the gauge weights.

Proof. Because the derivative $D_cF(c_*; R_0, \omega_0)$ is invertible, the implicit function theorem applies to the stationarity equation $F(c; R, \omega) = 0$. Hence there exist neighborhoods $\mathcal{U}$ and $\mathcal{V}$ and a unique C^1 branch $(R, \omega) \mapsto c(R, \omega)$ of stationary solutions with $c(R_0, \omega_0) = c_*$.

The C^1 regularity of this branch implies local boundedness of its derivative on sufficiently small neighborhoods. Therefore, after shrinking $\mathcal{U}$ if necessary, the branch satisfies a local Lipschitz estimate of the stated form.

The openness of the admissible-chart condition and the positive Jacobian margin ensure that the whole branch remains inside the admissible regime for (R, ω) close enough to (R_0, ω_0). Thus the result is not merely a stability statement for unconstrained stationary parameters: it is a stability statement for admissible structural charts selected by the gauge. This is precisely local robustness in the penalized quadratic class. □

Corollary 1 (Closed local core of $V3$)**.** *In the penalized quadratic class, the theory $V3$ possesses a closed local mathematical core consisting of:*

- *the exact affine compatibility with $V2$;*
- *a locally identifiable nonlinear gauge;*
- *a locally robust stationary chart selection.*

Proof. The affine compatibility proposition identifies the low-order core, the local identifiability theorem gives local uniqueness, and the local robustness theorem gives stability of the selected branch. Together they close the local theory in the penalized quadratic class. □

8 Ontological Limit and Transition to the Multichart Regime

The nonlinear single-chart theory remains valid only while one chart is structurally sufficient.

Definition 14 (Sufficient single-chart regime)**.** *I say that a density belongs to the sufficient single-chart regime if there exists an admissible nonlinear chart for which:*

- *the structural error remains controlled;*
- *the Jacobian remains uniformly away from degeneration;*
- *the chart does not require effective folding or self-overlap;*
- *the residual does not display irreducible fragmentation;*
- *no multichart alternative yields a decisive advantage together with ontological evidence of fragmentation.*

Proposition 5 (Ontological limit of $V3$)**.** *If every admissible single chart fails through persistent Jacobian degeneration, necessary folding, irreducible residual fragmentation, or multichart advantage accompanied by explicit fragmentation evidence, then the density lies outside the proper regime of $V3$ and must be treated in a multichart theory.*

Proof. Each of the listed failures contradicts one of the defining conditions of the sufficient single-chart regime. Therefore the single-chart ontology is no longer structurally adequate. The appropriate structural language must then change from a single chart to a multichart representation. □

9 Computational Evidence

The numerical experiments are included here only as evidence for the closed local core and for the ontological reading of the transition. The aggregate comparison between $V2$ and $V3$ is reported in Table 1, the three-way comparison with the multichart proxy in Table 2, the hard robustness outcomes in Table 3, and the degraded observational validation in Table 4.

Table 1: Global multiseed comparison between $V2$ and $V3$ on the classical 10-scenario batch.

Quantity	Value
Scenarios	10
Seeds per scenario	16
Total cases	160
Mean temporal error, $V2$	2.822×10^{-1}
Mean temporal error, $V3$	2.643×10^{-1}
Mean temporal gain $V3/V2$	$1.288\times$
Mean final error, $V2$	6.439×10^{-1}
Mean final error, $V3$	6.118×10^{-1}
Mean final gain $V3/V2$	$1.205\times$
Temporal wins of $V3$	160/160
Final wins of $V3$	160/160

Table 2: Global multiseed comparison among $V2$, $V3$, and a two-component multichart proxy $V4$.

Quantity	Value
Mean temporal error $(V2, V3, V4)$	$(2.798, 2.621, 1.746) \times 10^{-1}$
Mean final error $(V2, V3, V4)$	$(6.393, 6.085, 3.951) \times 10^{-1}$
Mean gain $V3/V2$	$1.289\times$
Mean gain $V4/V3$	$1.440\times$
Mean gain $V4/V2$	$1.755\times$
Wins of $V3$ over $V2$	160/160
Temporal wins of $V4$ over $V3$	143/160
Final wins of $V4$ over $V3$	144/160
Transition rate $V3 \to V4$	57.50%

The multiseed comparison shows that the nonlinear single-chart regime improves systematically on the affine regime in the tested batch. The comparison with the multichart proxy shows that fitting power alone cannot decide ontology. The robustness suite shows that explicit residual fragmentation is necessary to make the transition to the multichart regime structurally meaningful. The degraded observational validation shows that the computational advantage of $V3$ persists beyond the clean synthetic pipeline, and that a richer finite-dimensional family can improve fidelity further, although with higher Jacobian pressure in some scenarios.

10 Conclusion

I presented $V3$ as the canonical nonlinear global-chart structural decomposition beyond the affine theory. The affine decomposition of [1] remains the exact low-order core. The nonlinear part is

Table 3: Aggregate outcomes of the hard robustness suite for $V3$.

Stress regime	Mean gain toward $V4$	Fragmentation	Transition signal
Noise	20.97%	1.787	39.44%
Irregular sampling	16.72%	1.804	46.11%
Outliers	15.50%	1.092	13.89%
Progressive multimodality	27.81%	1.943	52.22%

Table 4: Degraded observational validation with partial tracking, heteroscedastic noise, anisotropic measurement bias, and sparse outliers.

Quantity	Value
Scenarios	10
Seeds per scenario	8
Total cases	80
Mean temporal error, $V2$	3.255×10^{-1}
Mean temporal error, $V3$ quadratic	3.118×10^{-1}
Mean temporal error, $V3$ cubic	2.185×10^{-1}
Mean gain, quadratic over $V2$	$1.072\times$
Mean gain, cubic over $V2$	$1.433\times$
Mean gain, cubic over quadratic	$1.348\times$
Quadratic wins over $V2$	80/80
Cubic wins over $V2$	80/80
Cubic wins over quadratic	80/80

selected by a hybrid gauge, and the local mathematical theory is closed first in smooth admissible Banach families and then explicitly in finite-dimensional subclasses, with the penalized quadratic class as the first realization. In that setting, I proved local identifiability, local robustness, exact compatibility with the affine theory, and I equipped the regime with a reaction-transport action, a derived continuity equation with source, and a corresponding structural distance.

This closes a precise stage of the theory: the maximal single-chart regime that can be treated locally in a mathematically controlled and computationally operational way. I do not claim here that the full nonlinear theory is closed in general nonlinear classes. The genuinely infinite-dimensional closure is now formulated at the Banach-space level under local differentiability, coercivity, and isomorphism hypotheses. What remains future work is to verify those hypotheses in concrete nonlinear families beyond the present explicit subclasses. The degraded observational validation nevertheless shows that the present framework is already informative outside the cleanest synthetic setting.

References

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Declarations

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Conflicts of interest I declare that there is no conflict of interest.

Data availability No external datasets were analyzed in the present version of this study.

Code availability The computational material used in the present manuscript is provided with the manuscript source, including the scripts `V3/quadratic_functional.py`, `V3/robustness_suite.py`, `V3/compare_v2_v3_batch10_multiseed.py`, and `V3/compare_v2_v3_v4_batch10_multiseed.py`, and is made available under the GNU AGPLv3 license in the project repository.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.